How-To Library

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Use Column Profile Results to Generate Test Data

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Abstract

In Test Data Manager, you can create a column profile and extrapolate the profile results to generate test data. TDM analyzes column data and infers rules based on the data. This article describes how to create and run a column profile, assign auto rules to target columns, and run an ad hoc generation plan to generate test data.

Supported Versions

- Test Data Management 9.7.1

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Overview

Create and run a column profile to determine the target column characteristics, such as a list of values, data patterns, and value ranges. Review the profile results, and extrapolate the profile results to generate test data.

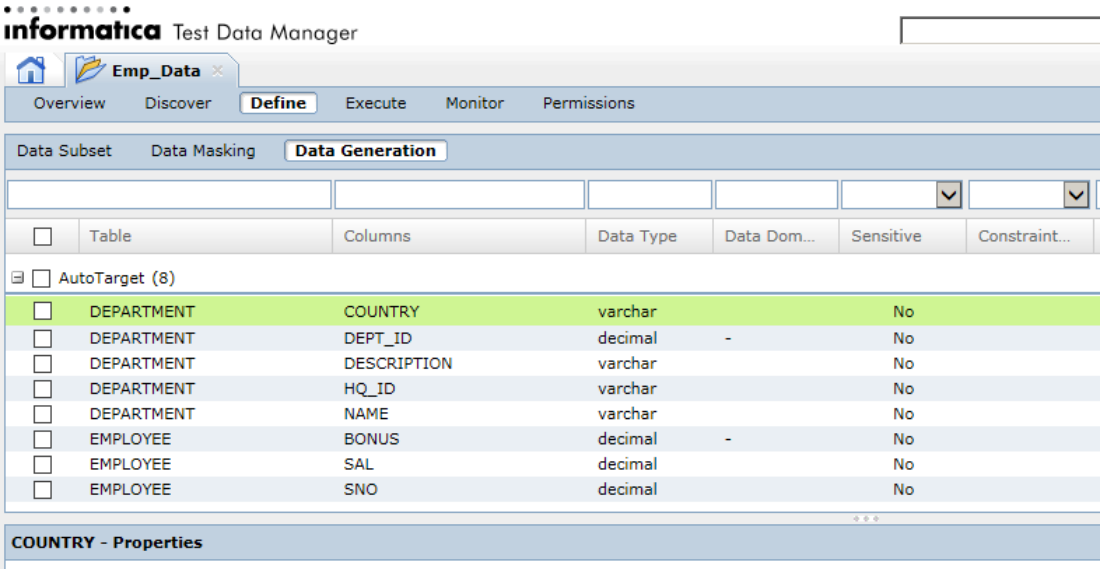
After you configure and run a column profile, TDM analyzes column data and infers rules based on the data.

Auto rules are the inferred rules that you assign after you run a column profile. When the status of inferred rule is Yes, you can assign auto rules to target columns to generate test data.

Scenario

You work for an organization in the software testing department. You need to test an application that maintains employee details such as employee IDs, salary, and bonus, and department details such as department IDs, names, and countries. To test this application, you need to generate test data. Create and run a column profile on the metadata, and use the profile results to generate test data.

The following image shows the metadata that you import from an Oracle database connection:



To generate test data, perform the following tasks:

1. Create a column profile.
2. Apply the column profile results.
3. Assign auto rules to target columns.
4. Generate and run an ad hoc plan.

Prerequisites

Perform the following prerequisites tasks:

- Install TDM.
- Create an Oracle database connection.
- Create a project, and import target metadata from the Oracle database connection.

Step 1. Create a Column Profile

Create and run a column profile to determine the inferred rules that you can apply to the target columns.

1. Open a project and click the **Discover** view.
2. Click the **Profile** view.

   The **Profile** view shows a list of the profiles in the project.
3. Click **Actions** > **New Profile**.

   The **New Profile** dialog box appears.
4. Enter a profile name. Optionally, enter a description.
5. Select **Column**.
6. Click **Next**.
7. To add the tables that you want to run a profile on, click **Add**.

   The **Select Tables** dialog box appears.

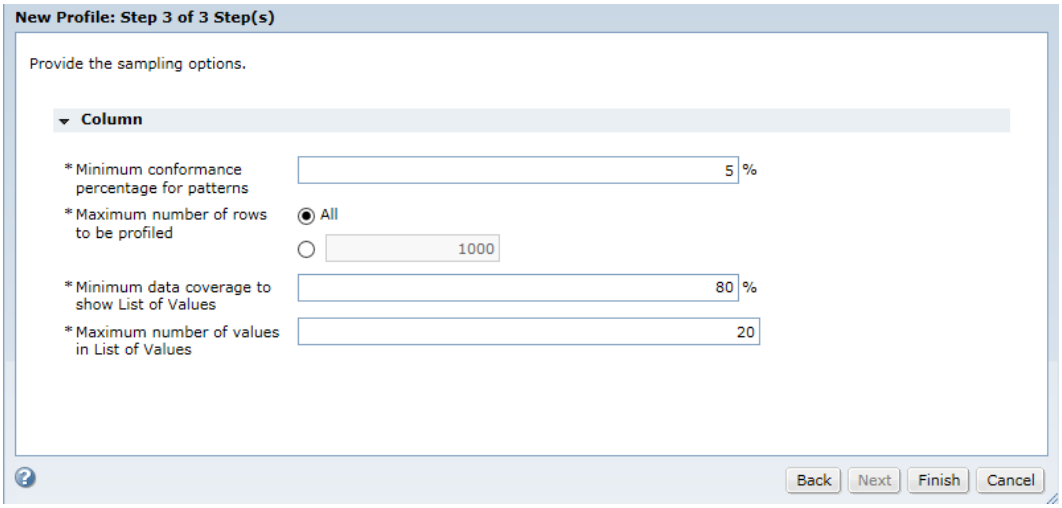
8. Select the tables, and click **OK**.

   The list of selected tables appears.

9. Click **Next**.

10. To create a column profile, you can configure the sampling options or use the default values.

    The following image shows the sampling options with the default values in a column profile:



11. Click **Finish**.

    The profile opens in a tab.

12. To run the profile, click **Actions** > **Execute**.

    The **Execute Profile** dialog box appears.

13. Select a connection to run the column profile.

14. Click **Execute**, and close the profile.

15. In the project, click **Monitor** to view the progress of the profile run.

Step 2. Apply the Column Profile Results

After you run a column profile, you can review the results and either accept or reject the results.

1. From the **Discover** | **Profile** view, open the profile that you created.

2. Click the **Column** tab.

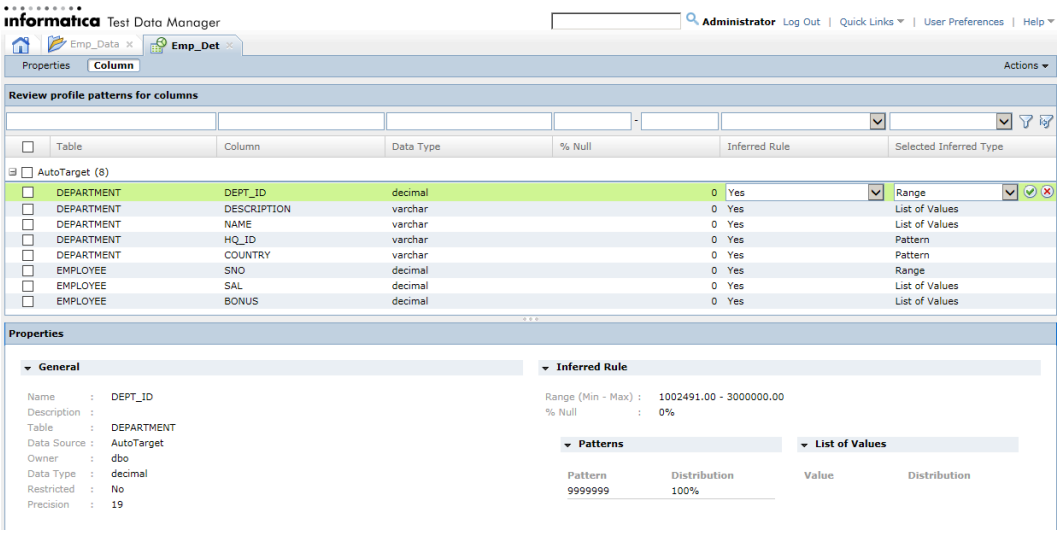
3. In the Inferred Rules column, select **Yes** for each target column to apply the inferred rule type.

4. In the Select Inferred Type column, select an inferred rule that you want to apply.

   The inferred rules are available based on the data type of the target column.

5. Click **Save**.

The following image shows a sample list of tables on which you can apply inferred rules:



6. Review and apply inferred rules for all the columns. Close the profile.

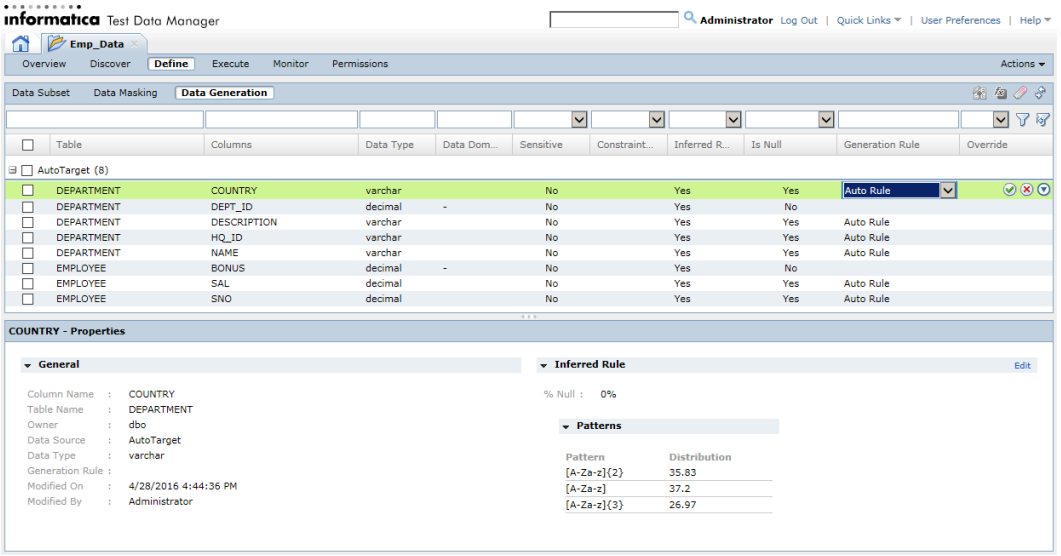
Step 3. Assign Auto Rules to Target Columns

You can assign an auto rule to target columns if the Inferred Rule status is Yes. You can edit the inferred rule properties after you assign the auto rule to the target.

1. In the project, click **Define** > **Data Generation**.
2. To assign auto rules to multiple columns, select all the rows that you want, and click **Generation Rule Assignment**.

   The **Apply Generation Rules** dialog box appears.
3. Select **Auto Rule**, and click **Apply** to save the assignment.

   The following image shows the list of target columns that contains auto rule assignments:



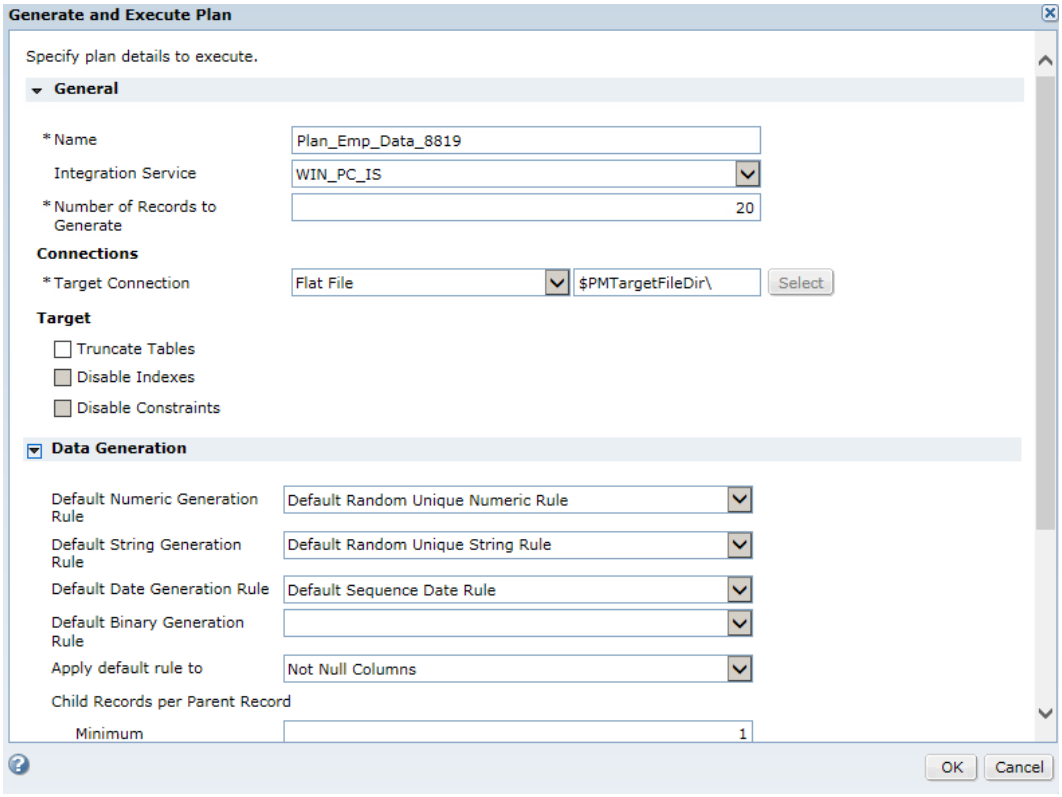
Step 4. Generate and Run an Ad Hoc Plan

Create an ad hoc generation plan for multiple target columns to generate test data.

1. Click **Define** > **Data Generation**.
2. Select all the target columns from the tables for which you want to generate test data.
3. Click **Actions** > **Generate and Execute**.

   The **Generate and Execute Plan** dialog box appears.
4. Enter a name for the plan or keep the default plan name that appears on the screen.
5. Select the PowerCenter Integration Service.
6. Enter the number of records that you want to generate.
7. Select a target connection.

   The following image shows the ad hoc plan properties that you can configure:

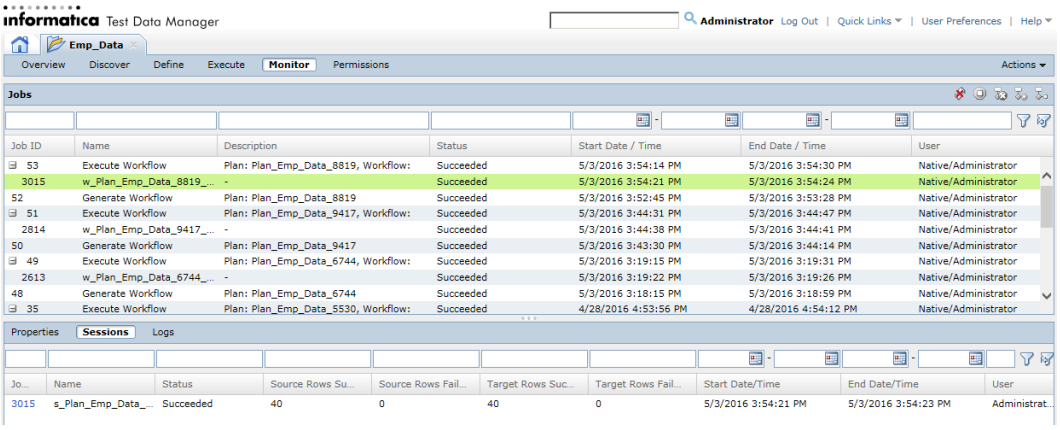


8. Select the default generation rules for the columns that do not contain any rule assignment.
9. Click **OK**.

   A **Create and Execute** dialog box appears.
10. To run the plan, click **Yes**.

11. To view the progress, click **Monitor**.

    The following image shows the progress of the data generation operation:



Sample Output

After the data generation plan succeeds, you can view the generated test data in the target.

The following images show the sample Employee and Department target files with 20 records each:

DEPARTMENT.out - Notepad

File Edit Format View Help

```
#DEPT_ID|NAME|DESCRIPTION|HQ_ID|COUNTRY
145|Software Dev|Development|ksE|USA
132|Software QA|Quality Analysis|tVR|USA
111|Hardware Testing|Testing|hyiK|Russia
108|Hardware Support|Support|li|India
102|Software Support|Support|DJ|UAE
103|Software Maintenance|Maintenance|S|China
167|Support Quality|Quality|obSR|USA
173|Hardware Maintenance|Maintenance|EvH|USA
125|Software Analytics|Analytics|D|India
113|Global Cust Support|Support|UhlX|Australia
119|Human Resources|HR|ZS|Australia
136|Others|Others|II|China
121|HW Tools|Hardware|J|India
178|SW Tools|Software|bjn|Russia
113|Hardware|Jade|IV|India
186|Finance|Financial|W|UK
110|Accounting|Accounts|wiCq|UK
192|Customer Care|Support|tq|UAE
151|Recruitment|HR|x|Japan
150|Billing|BillsJack|W|Japan
```

EMPLOYEE.out - Notepad

File Edit Format View Help

```
#SNO|SAL|BONUS
28|450000|2000
17|450000|2500
17|150000|4000
13|300000|3500
74|150000|1500
92|100000|3500
61|350000|500
30|300000|1500
38|350000|1000
43|550000|500
31|100000|4000
58|400000|5000
47|400000|3000
23|550000|2000
49|250000|1000
85|500000|5000
29|200000|4500
26|250000|4500
79|500000|3000
37|200000|2500
```

Author

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